Pontifical University
CATHOLIC UNIVERSITY OF PERU

FACULTY OF SOCIAL SCIENCES



Final Project

Course: Fundamentals of Econometrics
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Introduction

Childhood anemia is one of the most severe and persistent health problems in Peru. It can appear in the first years of life and affect brain development and long-term physical growth, leaving irreversible consequences. Despite the Peruvian government's ongoing health campaigns, the number of anemia cases and its prevalence remain alarmingly high. This suggests that the problem is not entirely medical, but rather linked to factors such as living conditions, poverty, the education level of caregivers, and the country's geographical inequalities. Therefore, this report aims to identify and analyze the main sociodemographic factors associated with the risk of anemia in children in Peru.

We will use official data from the Demographic and Family Health Survey (ENDES). We will propose an econometric model that considers the complexity of the chosen sample design, and we will evaluate how certain variables influence this risk with a simple but rigorous qualitative analysis, based on and comparing with what has been presented in the academic and econometric literature.

1. Literature review

According to Shimanda et al. (2020), these are the explanatory variables found with statistical significance and their expected effects on anemia:

- Child's age: This has a negative effect, as the risk of anemia decreases with age. This is because older children have lower iron requirements per kilogram of body weight and begin to consume more varied, abundant, and nutrient-rich foods. Age ranges were established, using the youngest group (6-11 months) as a reference point and the oldest group as a reference point, and a significant reduction was observed starting at 24 months.

- Child's sex: Indicates that anemia is significantly more common in male children (group (reference) than in girls. This is attributed to their higher rates of rapid growth, which increase their body's demand for iron, and not just any diet can satisfy that demand.

- Household wealth or well-being quintile: This is considered a determining factor in health. It has a negative effect, as belonging to households in lower income quintiles (poor and poorest) considerably increases the risk of developing anemia compared to the wealthiest quintile (reference group). This is explained by the severe economic limitations in accessing high-quality, nutritious food. However, an important finding of the study is that household wealth was only statistically significant for boys, while for girls, no significant association between quintile and anemia could be found.

These are the explanatory variables found to be statistically insignificant and their expected effects according to the literature:

- Place of residence: No relevant change was found in the risk of anemia between living in rural areas or in urban areas (reference group).

- Mother's educational level: No statistical significance was found for these levels of education, taking secondary education as a reference.

- Vitamin A supplementation: No significant effect was found between receiving supplementation or not with respect to anemia.

- Deworming medications: No significant effect was found between receiving this medication or not with respect to anemia.

- Child nutritional status: No significant effect was found between any of the three malnutrition indicators.

On the other hand, according to Velásquez-Hurtado et al. (2016), 12 independent factors associated with anemia in children under three years of age in Peru are presented. Logistic regression was used for both the univariate and multivariate analyses, calculating the Odds Ratios (OR). These are calculated as the ratio between the probability of an event occurring and the probability of it not occurring. They can be interpreted as the number of times more likely an event is to occur than not to occur. Odds ratios (OR) above 1 correspond to positive probit/logit coefficients, while those below 0 correspond to negative logit/probit coefficients. Thus, the respective odds ratios (OR) are obtained, taking 1 as the reference group (OR=1) with a 95% confidence interval. An OR=1 indicates that the variable increases the relative probability of presenting anemia compared to the reference category, while an OR<1 would indicate a protective effect.

- Region of residence: Children from the highlands had odds of suffering from anemia 1.3 times higher than those who resided in Lima and Callao, while those from the jungle had a probability 1.6 times higher.

- Quintile of well-being or household wealth: Children from households located in the first quintiles of wealth were found to have a higher odds of having anemia, being 2.1 times higher in the first two quintiles compared to the fifth quintile.

- Maternal age (years): When the mother was a teenager, the odds of children having anemia were 1.3 times higher than those whose mothers were not teenagers.

- Mother's level of education: When mothers had no level of education or onlyThey had a primary educationThe odds of children having anemia were 1.4 times higher than those children whose mothers had a higher level of education.

- Child's sex: In males, the odds of having anemia were 1.1 times higher compared to females.

- Child's age (months): Specifically, males under 12 months of age had odds of having anemia 4.4 times higher than those over 24 months.

- Fever in the child in the previous two weeks: When children had had a fever in the previous two weeks, the odds of having anemia were 1.1 times higher compared to children who had not suffered from this disease in the previous two weeks.

- Prenatal care in the first trimester: When there was no prenatal care in the first trimester of pregnancy, the odds of children having anemia were 1.1 times higher than those who did have such care in the first trimester of pregnancy.

- Childbirth in a health institution: When births were not attended in a health institution, the odds of children having anemia were 1.2 times higher compared to those children whose births were attended in a health institution.

- Iron supplementation during pregnancy (months): When no iron supplementation was received during pregnancy, the odds of children having anemia were 1.3 times higher compared to those who received a supplement administered for at least 6 months.

- Maternal anemia (Hb<12 g/dl): When the mother had a diagnosis of anemia at the time of the survey, the odds of children having anemia were 1.8 times higher compared to those children whose mothers did not have such a diagnosis at the time of the survey.

- Antiparasitic drugs for children: Children who did not receive intestinal antiparasitic treatment had a 1.3 times greater odds of having anemia than those who did receive such treatment.

Lastly, andBased on Fentaw et al. (2023), factors associated with anemia in children aged 6 to 59 months were analyzed using a multivariate logistic regression model. To measure the effect size of the

explanatory variables and isolate confounding factors, Adjusted Odds Ratios (AORs) were calculated with a 95% confidence interval. Variables with a p-value less than or equal to 0.05 are detailed below, along with their expected effects:

- Child's age: has a negative effect on the risk of anemia. The older the child, the lower the likelihood of developing anemia. Compared to the older control group (48 to 59 months), infants aged 6 to 11 months are 6.23 times more likely to have anemia, while the 12- to 23-month-old group is 3.74 times more likely. This is due to difficulties in transitioning to and consuming complementary foods after six months, which can lead to nutrient deficiencies, unlike older children who are already able to eat a more varied diet.

- Maternal age: has a negative effect. Children of mothers aged 30 or older are 63 percent less likely to suffer from anemia compared to those of younger mothers (AOR of 0.37). The literature explains this by stating that older mothers accumulate more experience in childcare and feeding practices, which decreases the risk of malnutrition in their children.

- Exclusive breastfeeding (EBF): Infants who were exclusively breastfed until six months of age were 73 percent less likely to develop anemia compared to those who were not (AOR of 0.27). This is based on the fact that EBF prevents respiratory infections and chronic diarrheal diseases, thus optimizing the infant's overall health.

- Dietary diversity score: Low consumption of varied foods is directly associated with anemia. Children with a low dietary diversity score are 2.61 times more likely to be anemic than those with an adequate diet. Consuming a wide variety of food groups ensures the intake of necessary nutrients and stimulates the child's immune system.

- History of diarrhea: Children who experienced episodes of diarrhea in the two months prior to assessment showed almost twice the risk (AOR of 1.87) of having anemia compared to those who did not become ill. Nutritional and fluid losses caused by diarrheal infections precipitate low hemoglobin levels.

- Monthly family income: This proves to be a critical socioeconomic factor. Households in the lowest income bracket (less than 1,500) drastically increase their risk, making children approximately 17 times more likely (AOR of 16.97) to develop anemia compared to families in the highest income bracket (greater than 4,500). Limited resources restrict food security and access to highly nutritious foods.

Furthermore, there is one explanatory variable without statistical significance in the multivariate model:

- Place of residence: Although the initial bivariate analysis showed a higher risk in rural areas, when adjusting the final multivariate model, no statistically significant association was found between the risk of anemia and residing in rural or urban areas (with a p-value equal to 0.119).

Because the SPSS commands used in the study were omitted, it is not possible to determine whether the effects correspond to marginal effects on average or on the average. Therefore, the results are interpreted strictly as Adjusted Odds Ratios (AOR), representing changes in the relative advantage (odds) and not direct variations in the absolute probability.

2. Model formulation and expected effects

To estimate the factors associated with anemia in children under three years of age in Peru, a binary response logit model was developed using the ENDES database. The dependent variable (called

“hw57”) originally identified children with no anemia, mild anemia, moderate anemia, or severe anemia. Therefore, it was modified to a dummy variable.

$$\left[Anemia_i = \begin{cases} 1 \text{ si el niño o niña presenta anemia severa, moderada o leve} \\ 0 \text{ si el niño o niña no presenta anemia} \end{cases} \right]$$

The proposed Logit model is expressed as:

$$P(Anemia_i = 1 \mid X_i) = \Lambda(X_i'\beta)$$

Where:

$$\Lambda(X_i'\beta) = \frac{e^{X_i'\beta}}{1 + e^{X_i'\beta}}$$

X*i*This represents the set of explanatory variables selected from the reviewed literature and the available information in the ENDES database. The proposed functional form is:

$$\begin{aligned} P(Anemia_i = 1 \mid X_i) = \Lambda(&\beta_0 + \beta_1 Hombre_i + \beta_2 Edad12a23_i + \beta_3 Fiebre_i + \beta_4 Diarrea_i \\ &+ \beta_5 DiversidadAlimentaria_i + \beta_6 LactanciaAntes12_i + \beta_7 MadreAdolescente_i \\ &+ \beta_8 EducNingPrim_i + \beta_9 EducSecundaria_i \\ &+ \beta_{10} AnemiaMadre_i + \beta_{11} HierroEmbarazo_i \\ &+ \beta_{12} RestoCosta_i + \beta_{13} Sierra_i + \beta_{14} Selva_i \\ &+ \beta_{15} Q1_i + \beta_{16} Q2_i + \beta_{17} Q3_i + \beta_{18} Q4_i) \end{aligned}$$

The selection of variables is not based solely on the statistical significance found in a single study, but rather on five criteria: relevance to the Peruvian case, consistency of the empirical evidence, theoretical plausibility, availability in the database, and the need to avoid including variables that measure very similar dimensions. The study by Velásquez-Hurtado et al. (2016) is the primary reference because it analyzes anemia in children aged 6 to 35 months using the ENDES survey from Peru and employs multivariate logistic regression. In this study, anemia is associated with sociodemographic factors, child characteristics, and maternal and child care variables, including region of residence, wealth quintile, maternal age and education, child's sex and age, recent fever, iron supplementation during pregnancy, maternal anemia, and antiparasitic treatment.

International evidence complements this selection. Fentaw et al. (2023) found that childhood anemia is associated with younger child age, low dietary diversity, recent diarrhea, and lower family income, while older maternal age and better feeding practices are associated with a lower risk of anemia. Furthermore, the study by Shimanda et al. (2020) shows that younger children, boys, and children from poorer households have a higher risk of anemia, although it did not find robust associations for some variables such as vitamin A, deworming, or anthropometric indicators.

Therefore, the main variables included are those supported by the literature and available in the database: child's sex, child's age, fever, diarrhea, dietary diversity, maternal age, maternal education, maternal anemia, iron supplementation during pregnancy, area of residence, natural region, household wealth quintile, and breastfeeding duration. These variables are then transformed for interpretation (See Appendix 1).

Based on the reviewed papers, sIt is expected that the boyshaveBoys are more likely to have anemia than girls, due to their higher iron requirements associated with growth. This is also expected in boys

aged 12 to 23 months.haveChildren aged 24 to 35 months are more likely to have anemia than those aged 24 to 35 months, as the literature shows that the risk is higher at younger ages, especially during the transition to complementary feeding. Recent episodes of fever and diarrhea should be positively associated with anemia, as they reflect infectious conditions that can affect nutritional status and micronutrient absorption.

Adequate dietary diversity should reduce the likelihood of anemia, because a more varied diet increases the likelihood of consuming foods rich in iron and other micronutrients.

The expected effect of thevariable lactationThe period before the first year is ambiguous. On the one hand, a shorter durationbreastfeedingIt could be associated with a higher likelihood of anemia if it reflects an early interruption of breastfeeding. However, it could also be associated with a lower likelihood of anemia if prolonged breastfeeding replaces adequate, iron-rich complementary feeding after six months. Therefore, the expected outcome is not conclusive and will depend on whether or not breastfeeding was accompanied by sufficient complementary feeding. Maternal education is also expected to have a negative effect on anemia, as more educated mothers may have better practices regarding nutrition, hygiene, prevention, and use of health services. Conversely, adolescent motherhood and maternal anemia should increase the likelihood of childhood anemia. Iron supplementation during pregnancy should reduce this likelihood, as it can improve the child's iron stores at birth.

Regarding household variables, living outside Metropolitan Lima is expected to increase the likelihood of anemia, especially in the Sierra and Selva regions. These variables reflect territorial differences in access to services, living conditions, nutrition, and healthcare. Similarly, children from households in the lowest income quintiles (Q1, Q2, Q3, and Q4) are expected to have a higher probability of anemia than those in the wealthiest quintile, because poverty limits access to nutritious food, basic services, and timely care.

Some variables present in the literature are not included in the main specification. Vitamin A, although available in the database, is not included as a central variable because its association is not robust in the reviewed multivariate models. It may be used later in a supplementary specification. Antiparasitic medication is relevant in the Peruvian study in the literature, but it is not included because an equivalent direct variable was not identified in the provided database. Area of residence is not included in the main model due to its potential collinearity with natural region and wealth quintile. The study that used the ENDES (National Demographic and Health Survey), although rural residence was initially significant, was excluded from the final model due to high collinearity. The model prioritizes natural region dummy variables, as they allow for capturing more specific territorial differences between Metropolitan Lima, the rest of the coast, the highlands, and the jungle. Finally, child nutritional status can be evaluated as an additional control or robustness test, but it is not included in the main model because it may be determined simultaneously with anemia due to common conditions of nutrition, health, and poverty.

3. Results

The model was run using Stata software, and marginal effects were applied to the average of the results. Additionally, the ENDES sample design was used: stratification and sample weights were included using the "svy" command (see Appendix 2). First, the nominal variables are presented. Regarding wealth quintiles, the highest quintile was used as the reference group. The results show that if the respondent belongs to the third quintile, the probability of her child having anemia increases by 11.69%; if she belongs to the second quintile, it is 12.97% more likely; and if she belongs to the first quintile, the probability increases by 9.4%. As for the mother's years of education, the reference group was those with higher education. Furthermore, a mother with secondary education increases the probability of her child having anemia by 8.1% compared to one with higher education. The last

nominal variable shows that belonging to the Sierra region increases the probability by 13% compared to living in Metropolitan Lima.

Likewise, if the mother is over 30 years old, the probability of anemia decreases by 3.9%. If the mother breastfeeds her child during the first year, the child's probability of having anemia is reduced by 5.9% compared to a mother who breastfeeds for a longer period. However, if the mother has anemia, the probability increases dramatically by 14.45%. Iron supplement use during pregnancy was not found to be significant. Regarding infants, being female is associated with a 5.2% reduction in the probability of having anemia; if they are over two years old, the probability is reduced by 22.39%; and if they have a varied diet (consuming more than four relevant food groups), the probability is also reduced by 3.3%. However, when the child has diarrhea or fever in the last two weeks, the probability of having anemia increases by 3.6% (if they had diarrhea) and by 2.9% (if they had fever).

4. Evaluation of model fit

In Logit-type models, it is not possible to use the traditional coefficient of determination***R^2***of the linear models estimated by Ordinary Least Squares (OLS). Furthermore, this model explicitly incorporates the complex sample design of the ENDES, which includes weighting, stratification, and clustering factors, thus breaking with the assumption of independent and identically distributed observations of the linear model. Therefore, classical statistics based on strict maximum likelihood (such as McFadden's Pseudo-R^2 or the Likelihood Ratio [LR] test) are not valid. To determine the validity and predictive power of our estimated model for childhood anemia, three essential dimensions adjusted for complex samples are evaluated: overall significance, calibration of specification, and correct classification ability (confusion matrix).

The first indicator is given by the Wald test adjusted with the F statistic: This test tests the null hypothesis that all the coefficients of the model are jointly equal to zero (H0: beta1= … = betak = 0). After running the regression, an F statistic (18,2981)=27.32 was obtained with a p-value of 0.000, so the null hypothesis is rejected and the model is considered globally significant at first.

The second indicator is obtained using the Archer and Lemeshow Goodness-of-Fit Test: Econometric literature recommends calculating the Archer and Lemeshow Test for complex samples like the one in this model. This assessment ranks individuals according totheir estimated probabilities and divides them into risk decilesWithin each decile, the test does not average the probabilities, but rather contrasts, using a Wald test, the distance between cases of anemia.observed(real) and theexpected(predicted by the model). The null hypothesis H0 indicates that the model is correctly specified (good fit); hypothesis H1 indicates that the model has specification problems (poor fit). After executing the command*state of being*The Archer-Lemeshow statistic yields an F-value of (9, 2990) with a p-value of 0.8356. Since the p-value is greater than 0.05, the null hypothesis cannot be rejected. Therefore, it is concluded that the calibrated model adequately fits the actual data distribution and does not suffer from severe problems of omitted variables or incorrect functional forms.

Finally, a third indicator is obtained from the confusion matrix.Using a standard cutoff threshold of 0.5 (50%(probability), the confusion matrix was constructed taking into consideration the weighting factors of the complex design. The results yieldedThe following are: in terms ofSensitivityThe model correctly identifies 45.33% of children who actually suffer from anemia (True Positives). This indicates that the model struggles to capture all positive cases because children's health depends on such complex and varied factors, which are difficult to capture in cross-sectional surveys. On the other hand, in terms ofSpecificityThe model excels at identifying healthy infants, achieving a specificity of 80.19% (True Negatives). Therefore, it is considered efficient for discriminating and excluding children who do not present a biological or socioeconomic risk of contracting the disease. Finally, theCounting R^2The calculated value is 66.08%, which, according to the expected parameters (range of 60-70%), indicates that the set of explanatory variables selected from the ENDES data, together

with our model, achieve a good fit andadequate, robust, and theory-consistent predictive powereconomic.

5. Discussion: Are there differentiated effects by area of residence?

Four explanatory variables were considered to differentiate the effects by area of residence (coast, highlands, and rainforest): wealth, maternal education, dietary variety, and maternal anemia. The aim was to evaluate the effect of these variables on the probability of childhood anemia according to area of residence. To this end, the interactions between each variable and area of residence were first assessed for statistical significance at the 5% level. However, based on the p-values, no significant evidence was found that the effect of the variables on the probability of childhood anemia differs according to area of residence, with the exception of maternal education. Regarding this last variable, the p-value was 0.0072, indicating significant evidence that the effect of maternal education on the probability of childhood anemia differs according to area of residence.

In general, it has been observed that the higher the mother's level of education, the lower the child's probability of having anemia. However, this effect varied depending on the area of residence. First, when the mother only had primary education, the estimated probability of the child having anemia was 45.74% higher in the highlands than on the coast. Second, when the mother reached a secondary education level, the estimated probability of the child having anemia remained higher in the highlands, in this case 51.31% higher than on the coast. Third, when the mother reached a higher education level, the estimated probability of the child having anemia was 43.68% higher in the highlands than on the coast. It remains consistent, then, that the higher the mother's level of education, the lower the estimated probability of the child having anemia, although with a greater effect in the highlands. In fact, when the mother had a higher education level, this was the only region where the probability was more than 40% higher compared to the coast. Furthermore, it was found that the effect of the mother's education level was much smaller in metropolitan Lima. Specifically, when the mother only had a primary education, the estimated probability of the child having anemia was 33.84% higher in metropolitan Lima than on the coast. Then, when the mother attained a secondary education level, the estimated probability of anemia was 39.08% higher in metropolitan Lima compared to the coast. Finally, when the mother attained a higher education level, the estimated probability of anemia was 31.98% higher in metropolitan Lima than on the coast. Therefore, a higher level of education would have increased the likelihood of anemia. This aligns with the centralization of educational quality in the country. In fact, the area of residence may not have had an inverse effect; that is, a higher level of maternal education may not have increased the likelihood of anemia. For example, a mother in metropolitan Lima, who has access to better opportunities, might invest a significant amount of time preparing for a good private university after completing secondary education, potentially neglecting her child. Conversely, a mother in the highlands who also completes secondary education, lacking opportunities to apply to university, will likely prefer to stay and work in any job to care for her child.

Literature

Fentaw, W., Belachew, T. y Andargie, A. (2023). Anemia and associated factors among 6 to 59 months age children attending health facilities in Kombolcha town, Northeast Ethiopia: a facility-based cross-sectional study. *BMC Pediatrics*, *23*(209), 1-9. https://doi.org/10.1186/s12887-023-04031-z

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years of age in Peru: analysis of the Demographic and Family Health Survey, 2007-2013.*Biomedical*, *36*(2), 220-229. https://doi.org/10.7705/biomedica.v36i2.2896

Attachments

Appendix 1: Explanatory variables

Variable	Construcción	Categoría de referencia	Efecto esperado
Sexo masculino	Dummy: 1 si es niño, 0 si es niña.	Niña	(+)
Edad de 12 a 23 meses	Dummy: 1 si tiene entre 12 y 23 meses, 0 si tiene entre 24 y 35 meses.	24 a 35 meses	(+)
Fiebre reciente	Dummy: 1 si tuvo fiebre en las últimas dos semanas, 0 si no tuvo fiebre.	Sin fiebre	(+)
Diarrea reciente	Dummy: 1 si tuvo diarrea en las últimas dos semanas, 0 si no tuvo diarrea.	Sin diarrea	(+)
Diversidad alimentaria	Dummy: 1 si presenta diversidad alimentaria adecuada, 0 si presenta baja diversidad alimentaria.	Baja diversidad alimentaria	(-)
Lactancia hasta antes del primer año	Dummy: 1 si lactó hasta antes del primer año, 0 si lactó después del primer año.	Lactancia después del primer año	(+/-)
Madre adolescente	Dummy: 1 si la madre tiene entre 15 y 19 años, 0 si tiene 20 años o más.	Madre no adolescente	(+)
Educación ninguna/primaria	Dummy: 1 si la madre no tiene educación o tiene primaria, 0 en otro caso.	Educación superior	(+)
Educación secundaria	Dummy: 1 si la madre tiene educación secundaria, 0 en otro caso.	Educación superior	(+)
Anemia materna	Dummy: 1 si la madre presenta anemia severa, moderada o leve; 0 si no presenta anemia.	Madre sin anemia	(+)
Hierro durante el embarazo	Dummy: 1 si recibió o compró hierro durante el embarazo, 0 si no recibió hierro.	No recibió hierro	(-)
Región natural	Dummies para Resto Costa, Sierra y Selva.	Lima Metropolitana	(+)
Quintil de riqueza	Dummies para Q1, Q2, Q3 y Q4.	Q5, quintil más rico	(+)

Original work

Appendix 2: Marginal effects of explanatory variables on anemia in children under 3 years of age in Peru

5 quantiles of wealth=1	0.094***	madre_30saños	-0.040***
	(0.029)		(0.015)
5 quantiles of wealth=2	0.130***	Sexo	-0.052***
	(0.028)		(0.015)
5 quantiles of wealth=3	0.117***	Edad_niño	-0.224***
	(0.025)		(0.015)
5 quantiles of wealth=4	0.034	Fiebre	0.030*
	(0.024)		(0.017)
Educación_madre= ninguno/primaria	0.022	Tiempo_lactancia	-0.059***
	(0.024)		(0.017)
Educación_madre= secundaria	0.081***	Hierro_suplemento	-0.033
	(0.020)		(0.024)
Residencia=Resto Costa	0.023	Diarrea	0.037**
	(0.021)		(0.019)
Residencia= Sierra	0.130***	Diversidad dietética	-0.033**
	(0.023)		(0.015)
Residencia=Selva	0.045*	Anemia_madre	0.145***
	(0.024)		(0.018)
		Observations	8667

* p < 0.10 ** p < 0.05 ***p < 0.01

Source: ENDES. Own elaboration

Appendix 3: Confusion matrix generated in Stata

```
Number of strata =     3                  Number of obs    =      8,667
Number of PSUs   = 3,001                  Population size = 4,669.9473
                                          Design df       =      2,998

------------------------------------------------------------------------
RECODE of hw57 |
(Nivel de      |
anemia en el   |
ni□o)          |  predicho_anemia
               |         0         1 |     Total
---------------+--------------------+----------
 no tiene      |      2230       551 |      2781
               |     .4774      .118 |     .5954
               |                     |
   anemia      |      1033     856.4 |      1889
               |     .2212     .1834 |     .4046
               |                     |
    Total      |      3263      1407 |      4670
               |     .6986     .3014 |         1
------------------------------------------------------------------------
Key: Weighted count
     Cell proportion

  Pearson:
    Uncorrected   chi2(1)        =  645.5163
    Design-based  F(1, 2998)     =  368.6613     P = 0.0000
```

Source: ENDES. Own elaboration

Appendix 4: Result of the margins of the interaction between the mother's level of education and the area of residence.

Columna1	b	se	t	pvalue	ll	ul	df	crit	eform
0bn.educació	0.250677	0.057152	4.386155	1.19E-05	0.138616	0.362738	2998	1.960756	0
0bn.educació	0.389919	0.026724	14.59042	1.24E-46	0.337519	0.442319	2998	1.960756	0
0bn.educació	0.445217	0.021369	20.83497	3.79E-90	0.403318	0.487116	2998	1.960756	0
0bn.educació	0.400061	0.024912	16.05917	9.77E-56	0.351215	0.448907	2998	1.960756	0
1.educación_	0.392585	0.023424	16.75984	2.46E-60	0.346656	0.438514	2998	1.960756	0
1.educación_	0.425935	0.016076	26.49463	3.7E-139	0.394414	0.457457	2998	1.960756	0
1.educación_	0.497013	0.017186	28.91994	2E-162	0.463316	0.53071	2998	1.960756	0
1.educación_	0.429723	0.019183	22.40146	6.8E-103	0.39211	0.467336	2998	1.960756	0
2.educación_	0.334168	0.028151	11.87057	8.58E-32	0.278971	0.389365	2998	1.960756	0
2.educación_	0.297562	0.019469	15.2841	7.67E-51	0.259388	0.335735	2998	1.960756	0
2.educación_	0.481622	0.023024	20.9186	8.22E-91	0.436479	0.526766	2998	1.960756	0
2.educación_	0.322211	0.025383	12.69405	5.3E-36	0.272442	0.371981	2998	1.960756	0

0bn.education: primary education; 1.education: secondary education; 2.education: higher education
Source: ENDES. Own elaboration

Appendix 5: Stata code for the report

```
*****************************************
* FUNDAMENTOS DE ECONOMETRÍA: Trabajo Grupal
* PROFESOR: Juan León Jara Almonte
*****************************************

clear all
cd "C:\Users\Camila\Downloads\grupo" // Introducir la ruta de acceso donde la
base de datos está ubicada
use "base_endes.dta", clear

*************************************************************
* 1) Transformación y selección de variables independientes
*************************************************************
des

* Ingreso: LISTO - IMPORTANTE
sum v191
gen wealth = v191/100000
sum wealth
xtile riqueza_quintil = wealth, nq(5)
tab riqueza_quintil

* Edad de la madre: LISTO - IMPORTANTE
sum v013
tab v013
tab v013, nolabel
recode v013 (1 2 3 = 0 "madres menores de 30 años") (4 5 6 7 = 1 "madres
mayores de 30 años"), gen(madre_mayor_30)
tab madre_mayor_30

* Sexo del niño: LISTO
tab sexo
tab sexo, nolabel
rename sexo mujer
tab mujer
```

```
* Edad del niño en meses: LISTO - IMPORTANTE
sum hw1
drop if hw1>36
sum hw1
egen edad_niño = cut(hw1), at(12, 24, 36) label
recode edad_niño (0 = 0 "menor a dos años")(1 = 1 "dos años o más"), gen (
edad_mayorigual2)
tab edad_mayorigual2

* Fiebre en las últimas dos semanas: LISTO
tab fiebre

* Tiempo de lactancia: LISTO - IMPORTANTE (Paper 3)
sum lactancia
tab lactancia
tab lactancia, nolabel
drop if lactancia == 0
recode lactancia (1 2 = 1 "lactancia hasta el primer año") (3 4 = 0 "lactancia
después del primer año"), gen (tiempo_lactancia)
tab tiempo_lactancia, nolabel

* Suplemento de hierro durante el embarazo: LISTO
tab m45
rename m45 hierro_embarazo
tab hierro_embarazo

* Diarrea en las últimas dos semanas: LISTO
tab diarrea

* Consumo variado de alimentos: LISTO - IMPORTANTE (Paper3)
tab diver_aliment
tab diver_aliment, nolabel
recode diver_aliment (0 1 2 3 4 = 0 "baja diversidad dietética") (5 6 7 = 1
"alta diversidad dietética"), gen(alta_vardietétitca)
tab alta_vardietétitca

* Anemia de la madre: LISTO
tab v457
tab v457, nolabel
recode v457 (4 = 0 "no tiene anemia") (1 2 3 = 1 "tiene anemia"), gen (
anemia_madre)
tab anemia_madre

* Años de escolaridad de la madre: LISTO - IMPORTANTE (Paper 2)
sum v133
tab v133
recode v133 (0 1 2 3 4 5 6 = 0 "ninguno/primaria") (7 8 9 10 11 = 1
"secundaria") (12 13 14 15 16 17 = 2 "superior"), gen (educación_madre)
tab educación_madre

* Anemia en niño: LISTO
tab hw57
tab hw57, nolabel
recode hw57 (1 2 3 = 1 "anemia") (4 = 0 "no tiene anemia"), gen (anemia_niño)
tab anemia_niño

* Lugar de residencia
tab shregion
tab shregion, nolabel
```

```
*****************
* 2) Modelo Logit
*****************

* Declarando el diseño muestral
svyset v001 [pweight = weight_m], strata (v022)

* Ejecutando el modelo logit
svy: logit anemia_niño ib5.riqueza_quintil ib2.educación_madre ib1.shregion
madre_mayor_30 mujer edad_mayorigual2 fiebre tiempo_lactancia hierro_embarazo
diarrea alta_vardietétitca anemia_madre

margins, dydx(*) atmeans
*** Riqueza: Grupo de referencia es el quintil más alto
*** Educación de la madre: El grupo de referencia es que la madre tiene
educación superior.
```

```
*****************
*3) Contrucción de los indicadores de la bondad del modelo
*****************

*Ver el F estadístico y su p-value

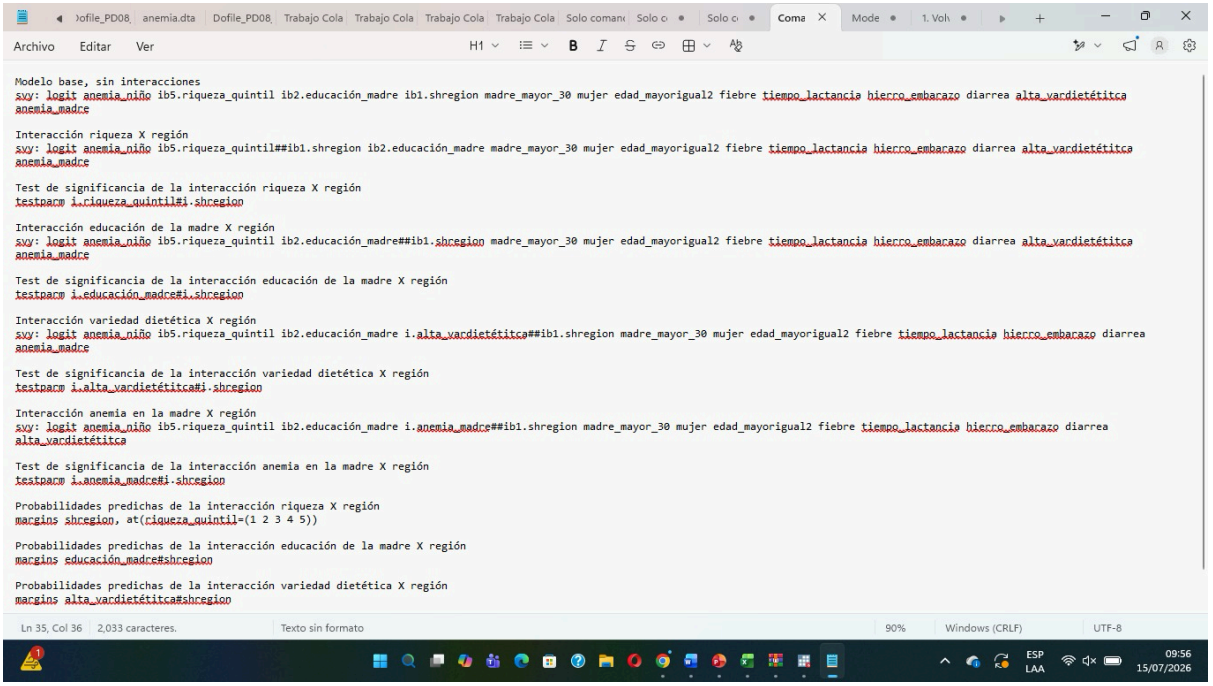
*Archer y Lemeshow
estat gof

*Matriz de confusión
predict prob_anemia if e(sample)

gen predicho_anemia = (prob_anemia >= 0.5) if prob_anemia != .

svy: tab anemia_niño predicho_anemia, count cell

**Cálculos
*A=2230, B=551, C=1033, D=856.4
***Sensibilidad
dis 856.4/(1033+856.4)
dis .45326559*100
*45.33%
***Especificidad
dis 2230/(2230+551)
dis .80186983*100
*80.19%
***R^2 de conteo
dis (2230+856.4)/(2230+551+1033+856.4)
dis .66084275*100
*66.08%
```


Archivo Editar Ver
Modelo base, sin interacciones
svy: logit anemia_niño ib5.riqueza_quintil ib2.educación_madre ib1.shregion madre_mayor_30 mujer edad_mayorigual2 fiebre tiempo_lactancia hierro_embarazo diarrea alta_vardietétitca anemia_madre
Interacción riqueza X región
svy: logit anemia_niño ib5.riqueza_quintil##ib1.shregion ib2.educación_madre madre_mayor_30 mujer edad_mayorigual2 fiebre tiempo_lactancia hierro_embarazo diarrea alta_vardietétitca anemia_madre
Test de significancia de la interacción riqueza X región
testparm i.riqueza_quintil#i.shregion
Interacción educación de la madre X región
svy: logit anemia_niño ib5.riqueza_quintil ib2.educación_madre##ib1.shregion madre_mayor_30 mujer edad_mayorigual2 fiebre tiempo_lactancia hierro_embarazo diarrea alta_vardietétitca anemia_madre
Test de significancia de la interacción educación de la madre X región
testparm i.educación_madre#i.shregion
Interacción variedad dietética X región
svy: logit anemia_niño ib5.riqueza_quintil ib2.educación_madre i.alta_vardietétitca##ib1.shregion madre_mayor_30 mujer edad_mayorigual2 fiebre tiempo_lactancia hierro_embarazo diarrea anemia_madre
Test de significancia de la interacción variedad dietética X región
testparm i.alta_vardietétitca#i.shregion
Interacción anemia en la madre X región
svy: logit anemia_niño ib5.riqueza_quintil ib2.educación_madre i.anemia_madre##ib1.shregion madre_mayor_30 mujer edad_mayorigual2 fiebre tiempo_lactancia hierro_embarazo diarrea alta_vardietétitca
Test de significancia de la interacción anemia en la madre X región
testparm i.anemia_madre#i.shregion
Probabilidades predichas de la interacción riqueza X región
margins shregion, at(riqueza_quintil=(1 2 3 4 5))
Probabilidades predichas de la interacción educación de la madre X región
margins educación_madre#shregion
Probabilidades predichas de la interacción variedad dietética X región
margins alta_vardietétitca#shregion
Ln 35, Col 36 2,033 caracteres. Texto sin formato 90% Windows (CRLF) UTF-8
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